GGG-T-60A
November 24, 1970
SUPERSEDING
Fed. Spec. GGG-T-60
November 1, 1956

FEDERAL SPECIFICATION

TAPS, THREAD CUTTING (CARBON STEEL, HAND, CUT THREAD, UNIFIED, AMERICAN AND METRIC THREAD FORM, SCREW AND BOLT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers carbon-steel taps of the Unified and American National thread series machine screw and fractional sizes.

1.1.1 Federal specification coverage. This specification does not include all types of the taps indicated by the title of this specification or which are commercially available, but is intended to cover only the type and series which are generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, series, and styles. The carbon-steel taps listed herein shall be of the following types, series, and styles, as specified (see 6.2):

Type I - Unified and American Thread form.
Series A - Coarse thread (symbol UNC).
Series B - Fine thread (symbol UNF).
Series C - Special thread (symbol UNS).

Type II - Metric thread form.
Series D - French standard.
Series E - International standard.
Series F - Spark plug sizes.
Series G - Special.

1.2.2 Styles. Taps shall be of the following styles, as specified (see 6.2):

Style 1 - Taper tap.
Style 2 - Plug tap.
Style 3 - Bottoming tap.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

FSC 5136

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau Of Standards (NBS) Handbook:
H-28, Part I - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

American National Standards Institute (ANSI) Standards:
B1.1 - Unified Screw Threads.
B94.9 - Taps, Cut and Ground Threads.
Z76.4 - Standard Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness, and Knoop Hardness) (ASTM E140)

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of taps otherwise conforming to this specification.

3.2 Manufacturer's current product. The taps furnished by the supplier shall be the manufacturer's highest quality unused standard current product that has already performed satisfactorily in industry.

3.3 Thread form. The Unified and American National form of thread for taps shall conform to H-28, Part I, and B1.1. The metric form of thread for taps shall conform to the French and International standards.

3.4 Material The carbon steel used in the taps shall be of such composition as to produce taps of the highest quality manufactured in the tool industry.

3.5 Marking.

3.5.1 Marking of taps. The taps shall be permanently marked with the manufacturer's name or with a trademark of such known character that the source of manufacture may be

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readily determined. The taps shall be marked with the nominal diameter, threads per inch, and thread symbol. Left hand threads shall be marked "L.H." The metric cut threads shall be marked with the nominal basic size and the pitch in millimeters.

3.6 Hand of thread. Unless otherwise specified (see 6.2), right hand cutting taps shall be furnished.

3.7 Tap centers. Cut-thread taps shall have suitable centers provided at thread end. Tap sizes 3/8 inch and smaller may have external centers on the thread end; sizes larger than 3/8 inch shall have internal centers on thread end. Cut thread tap sizes 1/4 inch through 3/8 inch may have external partial cone center on shank end, with the length of the cone center approximately 1/4 of diameter of shank. The metric cut taps shall have external centers on the thread end on sizes up to and including 8 mm. except that 7 and 8 mm. bottoming shall have internal centers, 9 mm. and larger shall have internal center on thread end.

3.8 Finish. All surfaces of the tap shall be smooth and finished in accordance with good commercial practice.

3.8.1 Threads. The threads (flanks and crests) shall have a smooth finish, free from scales, burrs, nicks, cutter, and chatter marks or other imperfections which may affect the accuracy of the tap.

3.8.2 Flutes. The straight flutes shall be smoothly ground after hardening to produce an unobstructed chip surface and to effect a refined cutting edge to the crest of the teeth. Flutes shall be accurately indexed and shall be of sufficient depth to provide ample chip clearance.

3.9 Heat treatment. The full length of the threaded end of the tap shall be suitably hardened (see 4.3.2) and tempered. The taps shall have a Rockwell hardness of not less than 61 nor more than 65 on the C scale when measured in the first 80 percent of the threaded section. The shanks and square shall have a minimum Rockwell hardness of C-45.

3.10 Shank. Shanks of series A through G-form taps shall be ground smoothly after hardening.

3.11 Eccentricity tolerances. Eccentricity tolerances of thread elements shall be as specified in table I.

TABLE I. Eccentricity tolerances of thread elements

Element	Range Sizes are Inclusive		Cut Thread	
	Hand Mch. Screw	Pipe	Eccen-tricity	t.i.v.[1/]
	Inches	Inches	Inch	Inch
Square (at central point)	0-1/2	1/16-1/8	.0030	.0060
	17/32-4	1/4-4	.0040	.0080
Shank	0-5/16	1/16	.0030	.0060
	11/32-4	1/8-4	.0040	.0080
Major Diameter	0-5/16	1/16	.0025	.0050
	11/32-4	1/8-4	.0040	.0080
Pitch Diameter (at first full thrd.)	0-5/16	1/16	.0025	.0050
	11/32-4	1/8-4	.0040	.0080
Chamfer[2/]	0-1/2	1/16-1/8	.0020	.0040
	17/32-4	1/4-4	.0030	.0060

[1/] t.i.v. = Total indicator variation. Figures are given for both eccentricity and total indicator variation to avoid misunderstanding.

[2/] Chamfer should preferably be inspected by light projection to avoid errors due to indicator contact points dropping into the thread grooves.

Note: Metric taps shall conform to the nearest inch size above.

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3.12 Chamfers. Chamfers shall be ground smoothly after hardening with a minimum of chatter marks.

3.12.1 Series A through G style 1, taper taps. Taper taps shall be chamfered on the threaded end for a length of from 8 to 10 threads. Taper taps shall be similar in design to figure 1 for sizes 3/8 inch and smaller and figure 2 for sizes larger than 3/8 inch.

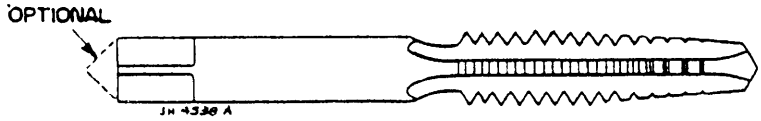


Figure 1. Style 1, taper tap, sizes 3/8 inch and smaller.

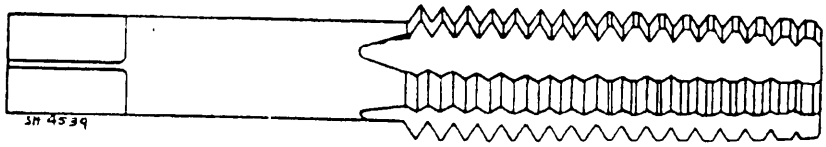


Figure 2. Style 1, taper tap, sizes larger than 3/8 inch.

3.12.2 Series A through series G, style 2, plug taps. Plug taps shall be chamfered on the threaded end for a distance of from 3 to 5 threads. Plug taps shall be similar in design to figure 3 for sizes smaller than 3/8 inch and figure 4 for sizes 3/8 inch and larger.

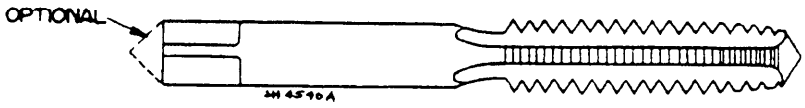


Figure 3. Style 2, plug tap, sizes 3/8 inch and smaller.

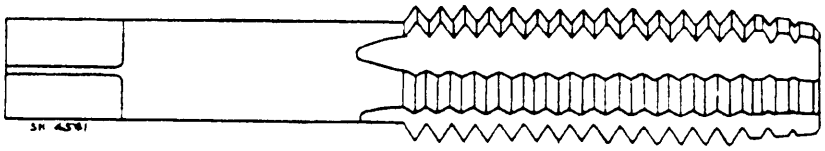


Figure 4. Style 2, plug tap, sizes larger than 3/8 inch.

3.12.3 Series A through G style 3, bottoming taps. Bottoming taps shall be chamfered on the threaded end for a length of from 1 to 1-1/2 threads. Bottoming taps shall be similar in design to figure 5 for sizes 3/8 inch and smaller and figure 6 for sizes larger than 3/8 inch.

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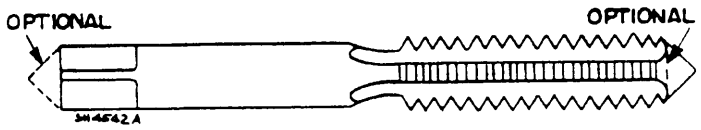


Figure 5. Style 3, bottoming tap, sizes 3/8 inch and smaller.

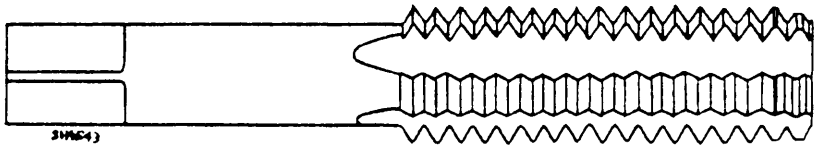


Figure 6. Style 3, bottoming tap, sizes larger than 3/8 inch.

3.13 Dimensions and tolerances.

3.13.1 The general dimensions for cut-thread hand taps, type I, series A through C shall conform to table II for the size specified. Tolerances shall be as specified in table III.

TABLE II. Dimensions for series A through C cut thread hand taps

Screw gage number or diameter, inches	Threads per inch			Number of flutes	Length overall	Length of thread	Length of square	Diameter of shank	Size of square
	Series A	Series B	Series C						
					Inches	Inches	Inches	Inches	Inches
0.....	.....	80	.....	2	1-5/8	5/16	3/16	0.141	0.110
1.....	64	72		2	1-11/16	3/8	3/16	.141	.110
2.....	56	64	.....	3	1-3/4	7/16	3/16	.141	.110
3.....	48	56	.....	3	1-13/16	1/2	3/16	.141	.110
4.....	40	48	32 36	3	1-7/8	9/16	3/16	.141	.110
5.....	40	44		3	1-15/16	5/8	3/16	.141	.110
6.....	32	40	36	3	2	11/16	3/16	.141	.110
8.....	32	36	40	4	2-1/8	3/4	1/4	.168	.131
10.....	24	32	30	4	2-3/8	7/8	1/4	.194	.152
12.....	24	28	32	4	2-3/8	15/16	9/32	.220	.165
14.....	.....	.....	20 24	4	2-1/2	1	5/16	.255	.191
1/ 1/16...	.....	.....	64	2	1-5/8	5/16	3/16	.141	.110
1/ 3/32...	.....	.....	48	3	1-13/16	1/2	3/16	.141	.110
1/ 1/8....	.....	.....	40	3	1-15/16	5/8	3/16	.141	.110
1/ 5/32...	.....	.....	32 36	4	2-1/8	3/4	1/4	.168	.131
1/ 3/16...	.....	.....	24 32	4	2-3/8	7/8	1/4	.194	.152
1/ 7/32...	.....	.....	24 32	4	2-3/8	15/16	9/32	.220	.165
1/4....	20	28	24 32	4	2-1/2	1	5/16	.255	.191
5/16...	18	24	32	4	2-23/32	1-1/8	3/8	.318	.238
3/8....	16	24	.....	4	2-15/16	1-1/4	7/16	.381	.286

1/ Whenever possible, the substitution of machine screw taps is recommended for these sizes.

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TABLE II. Dimensions for series A through C cut thread taps (continued)

Screw gage number or diameter, inches	Threads per inch			Number of flutes	Length overall	Length of thread	Length of square	Diameter of shank	Size of square
	Series A	Series B	Series C						
					Inches	Inches	Inches	Inches	Inches
7/16...	14	20		4	3-5/32	1-7/16	13/32	0.323	0.242
1/2....	13	20		4	3-3/8	1-21/32	7/16	.367	.275
9/16...	12	18	.....	4	3-19/32	1-21/32	1/2	.429	.322
5/8....	11	18	12	4	3-13/16	1-13/16	9/16	.480	.360
11/16...	.....	.....	11 16	4	4-1/32	1-13/16	5/8	.542	.406
3/4....	10	16		4	4-1/4	2	11/16	.590	.442
7/8....	9	14		4	4-11/16	2-7/32	3/4	.697	.523
1.....	8	12	14	4	5-1/8	2-1/2	13/16	.800	.600
1-1/8...	7	12	.....	4	5-7/16	2-9/16	7/8	.896	.672
1-1/4...	7	12		[2/] 4	5-3/4	2-9/16	1	1.021	.766
1-3/8...	6	12	.....	[2/] 4	6-1/16	3	1-1/16	1.108	.831
1-1/2...	6	12	.....	[2/] 4	6-3/8	3	1-1/8	1.233	.925
1-3/4	5	.....	.....	6	7	3-3/16	1-1/4	1.430	1.072
2.....	4-1/2	.....	.....	6	7-5/8	3-9/16	1-3/8	1.644	1.233

[1/] Wherever possible, the substitution of machine screw taps is recommended for these sizes.

[2/] Series B taps shall have six flutes in these sizes.

TABLE III. Tolerances of general dimensions for series A through C cut thread taps

Elements	Range	Tolerance
		Inch
Length overall	Nos. 0 to 14, incl.	±1/32
	1/16 to 1 inch, incl.	±1/32
	1-1/16 to 2 inches, incl.	±1/16
	Nos. 0 to 12, incl.	±3/64
	No. 14	±1/16
Length of thread	1/16 to 7/32 inch, incl.	±3/64
	1/4 to 1/2 inch, incl.	±1/16
	9/16 to 1-1/2 inches, incl.	±3/32
	1-5/8 to 2 inches, incl.	±1/8
Length of square	Nos. 0 to 14, incl.	±1/32
	1/16 to 1 inch, incl.	±1/32
	1-1/16 to 4 inches, incl.	±1/16
	Nos. 0 to 14 incl.	-0.005
	1/16 to 7/32 inch incl.	- .004
	1/4 to 1 inch. incl.	- .005
Diameter of shank	1-1/16 to 2 inches incl.	- .007
	Nos. 0 to 14, incl.	- .004
Size of square	1/16 to 1/2 inch, incl.	- .004
	9/16 to 1 inch, incl.	- .006
	1-1/16 to 2 inches, incl.	- .008

3.13.2 The dimensions for cut-thread hand taps, type II, series D through G shall conform to table IV for the size specified (see 6.2). The tolerances shall be as specified in table V.

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TABLE IV. Dimensions for type II, series D through G in millimeters

Diameter of tap	Pitch mm. Series D	Series E	Series F	Series G	Number of flutes	Length overall	Length of thread	Length of square	Diameter of shank	Size of square
Mm.						Mm.	Mm.	Mm.	Mm.	Mm.
1.5...	0.35	.....	.....	0.35	2	41.3	7.9	4.76	3.58	2.79
2.....	.45	.....	.....	.....	3	44.5	11.1	4.76	3.58	2.79
2.5...	.45	.....	.....	.....	3	46.0	12.7	4.76	3.58	2.79
3.....	.60	.....	.....	.....	3	49.2	15.9	4.76	3.58	2.79
3.5...	.60	0.60	.....	.....	3	50.8	17.5	4.76	3.58	2.79
4.....	.75	.....	.....	.....	4	54.0	19.1	6.35	4.27	3.33
4.5...	.75	.75	.....	.....	4	60.3	22.2	6.35	4.93	3.86
5.....	.90	.....	.....	.....	4	60.3	23.8	7.14	5.59	4.19
5.5...	.90	.90	.....	.....	4	60.3	23.8	7.14	5.59	4.19
6.....	1.00	1.00	.....	.....	4	63.5	25.4	7.14	6.48	4.85
7.....	1.00	1.00	.....	.....	4	69.1	28.6	7.94	8.08	6.05
8.....	.....	1.25	.....	.....	4	69.1	28.6	8.73	8.08	6.05
9.....	.....	1.25	.....	.....	4	74.6	31.8	8.73	6.99	5.23
10.....	1.50	.....	1.00	.....	4	74.6	31.8	9.52	7.77	5.82
12.....	.....	1.75	.....	.....	4	85.7	42.1	10.31	9.32	6.99
14.....	2.00	2.00	1.25	.....	4	91.3	42.1	12.70	10.90	8.18
16.....	2.00	2.00	.....	.....	4	96.8	46.0	14.29	12.19	9.14
18.....	2.50	2.50	1.50	.....	4	102.4	46.0	15.87	13.77	10.31
20.....	2.50	2.50	.....	.....	4	113.5	50.8	17.46	16.56	12.42
22.....	2.50	2.50	.....	.....	4	119.1	56.4	17.46	17.70	13.28
24.....	3.00	3.00	.....	.....	4	124.6	56.4	19.05	19.30	14.48

TABLE V. Tolerances of general dimension, type II, series D through G

Elements	Range	Tolerance
	Mm.	Mm.
Length overall	1.5 to 24, incl.	±0.79
	1.5 to 5.5, incl.	±1.19
Length of thread	6 to 12, incl.	±1.58
	14 to 24, incl.	±2.38
Length of square	1.5 to 24, incl.	± .79
Diameter of shank	1.5 to 5.5, incl.	- .10
	6 to 24, incl.	- .13
Size of square	1.5 to 12, incl.	- .10
	14 to 24, incl.	- .15

3.14 Thread limits for type I series A through C. Thread limits for the major and pitch diameters of series A through C shall conform to table VI (see 6.2).

3.15 Thread limits for type II series D through G. Thread limits for the major and pitch diameters of cut taps series D through G shall conform to table VII for the size specified (see 6.2).

3.16 Lead tolerance. A maximum lead error of plus or minus 0.003 inch in 1 inch of thread will be permitted for all cut threads.

3.17 Angle tolerance. The angle tolerances specified in table VIII for cut thread taps will be permitted.

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TABLE VI. Thread limits, type I, series A through C, cut thread taps

Screw gage number or diameter	Threads per inch			Major diameter limits			Pitch diameter limits		
	Series A	Series B	Series C	Basic	Min.	Max.	Basic	Min.	Max.
Inches				Inch	Inch	Inch	Inch	Inch	Inch
0.....	.....	80	.....	0.0600	0.0609	0.0624	0.0519	0.0521	0.0531
1.....	64	.....	.....	.0730	.0740	.0755	.0629	.0631	.0641
1.....	.....	72	.....	.0730	.0740	.0755	.0640	.0642	.0652
2.....	56	.....	.....	.0860	.0872	.0887	.0744	.0746	.0756
2.....	.....	64	.....	.0860	.0870	.0885	.0759	.0761	.0771
3.....	48	.....	.....	.0990	.1003	.1018	.0855	.0857	.0867
3.....	.....	56	.....	.0990	.1002	.1017	.0874	.0876	.0886
4.....	.....	.....	32	.1120	.1142	.1162	.0917	.0922	.0937
4.....	.....	.....	36	.1120	.1137	.1157	.0940	.0942	.0957
4.....	40	.....	.....	.1120	.1136	.1156	.0958	.0960	.0975
4.....	.....	48	.....	.1120	.1133	.1153	.0985	.0987	.1002
5.....	40	.....	.....	.1250	.1266	.1286	.1088	.1090	.1105
5.....	.....	44	.....	.1250	.1264	.1284	.1102	.1104	.1119
6.....	32	.....	.....	.1380	.1402	.1422	.1177	.1182	.1197
6.....	.....	.....	36	.1380	.1397	.1417	.1200	.1202	.1217
6.....	.....	40	.....	.1380	.1396	.1416	.1218	.1220	.1235
8.....	32	.....	.....	.1640	.1662	.1682	.1437	.1442	.1457
8.....	.....	36	.....	.1640	.1657	.1677	.1460	.1462	.1477
8.....	.....	.....	40	.1640	.1656	.1676	.1478	.1480	.1495
10.....	24	.....	.....	.1900	.1928	.1948	.1629	.1634	.1649
10.....	.....	.....	30	.1900	.1923	.1943	.1684	.1689	.1704
10.....	.....	32	.....	.1900	.1922	.1942	.1697	.1702	.1717
12.....	24	.....	.....	.2160	.2188	.2208	.1889	.1894	.1909
12.....	.....	28	.....	.2160	.2184	.2204	.1928	.1933	.1948
12.....	.....	.....	32	.2160	.2182	.2202	.1957	.1962	.1977
14.....	.....	.....	20	.2420	.2452	.2477	.2095	.2100	.2120
14.....	.....	.....	24	.2420	.2448	.2473	.2149	.2154	.2174
1/16..	.....	.....	64	.0625	.0635	.0650	.0524	.0526	.0536
3/32..	.....	.....	48	.0938	.0951	.0966	.0803	.0805	.0815
1/8...	.....	.....	40	.1250	.1266	.1286	.1088	.1090	.1105
5/32..	.....	.....	32	.1563	.1585	.1605	.1360	.1365	.1380
5/32..	.....	.....	36	.1563	.1580	.1600	.1382	.1384	.1399
3/16..	.....	.....	24	.1875	.1903	.1923	.1604	.1609	.1624
3/16..	.....	.....	32	.1875	.1897	.1917	.1672	.1677	.1692
7/32..	.....	.....	24	.2188	.2216	.2236	.1917	.1922	.1937
7/32..	.....	.....	32	.2188	.2210	.2230	.1985	.1990	.2005
1/4...	20	.....	.....	.2500	.2532	.2557	.2175	.2180	.2200
1/4...	.....	.....	24	.2500	.2528	.2553	.2229	.2234	.2254
1/4...	.....	28	.....	.2500	.2524	.2549	.2268	.2273	.2288
1/4...	.....	.....	32	.2500	.2522	.2547	.2297	.2302	.2317
5/16..	18	.....	.....	.3125	.3160	.3185	.2764	.2769	.2789
5/16..	.....	24	.....	.3125	.3153	.3178	.2854	.2859	.2874
5/16..	.....	.....	32	.3125	.3147	.3172	.2922	.2927	.2942
3/8...	16	.....	.....	.3750	.3789	.3814	.3344	.3349	.3369
3/8...	.....	24	.....	.3750	.3778	.3803	.3479	.3484	.3499
7/16..	14	.....	.....	.4375	.4419	.4449	.3911	.3916	.3941
7/16..	.....	20	.....	.4375	.4407	.4437	.4050	.4055	.4075
1/2...	13	.....	.....	.5000	.5047	.5077	.4500	.4505	.4530
1/2...	.....	20	.....	.5000	.5032	.5062	.4675	.4680	.4700
9/16..	12	.....	.....	.5625	.5675	.5705	.5084	.5089	.5114
9/16..	.....	18	.....	.5625	.5660	.5690	.5264	.5269	.5289
5/8...	11	.....	.....	.6250	.6304	.6334	.5660	.5665	.5690
5/8...	.....	.....	12	.6250	.6300	.6330	.5709	.5714	.5739
5/8...	.....	18	.....	.6250	.6285	.6315	.5889	.5894	.5914
11/16.	.....	.....	11	.6875	.6929	.6969	.6285	.6290	.6320
11/16.	.....	.....	16	.6875	.6914	.6954	.6469	.6474	.6499
3/4...	10	.....	.....	.7500	.7559	.7599	.6850	.6855	.6885
3/4...	.....	16	.....	.7500	.7539	.7579	.7094	.7099	.7124
7/8...	9	.....	.....	.8750	.8820	.8860	.8028	.8038	.8068
7/8...	.....	14	.....	.8750	.8799	.8839	.8286	.8296	.8321

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TABLE VI. Thread limits for series A through C cut thread taps (continued)

Screw gage number or diameter	Threads per inch			Major diameter limits			Pitch diameter limits		
	Series A	Series B	Series C	Basic	Min.	Max.	Basic	Min.	Max.
Inches				Inch	Inch	Inch	Inch	Inch	Inch
1.....	8	.....	.....	1.0000	1.0078	1.0118	.9188	.9198	.9228
1.....	.....	12	.....	1.0000	1.0055	1.0095	.9459	.9469	.9499
1.....	.....	.....	14	1.0000	1.0049	1.0089	.9536	.9546	.9571
1-1/8..	7	.....	.....	1.1250	1.1337	1.1382	1.0322	1.0332	1.0367
1-1/8..	.....	12	.....	1.1250	1.1305	1.1350	1.0709	1.0719	1.0749
1-1/4..	7	.....	.....	1.2500	1.2587	1.2632	1.1572	1.1582	1.1617
1-1/4..	.....	12	.....	1.2500	1.2555	1.2600	1.1959	1.1969	1.1999
1-3/8..	6	.....	.....	1.3750	1.3850	1.3895	1.2667	1.2677	1.2712
1-3/8..	.....	12	.....	1.3750	1.3805	1.3850	1.3209	1.3219	1.3249
1-1/2..	6	.....	.....	1.5000	1.5100	1.5145	1.3917	1.3927	1.3962
1-1/2..	.....	12	.....	1.5000	1.5055	1.5100	1.4459	1.4469	1.4499
1-3/4..	5	.....	.....	1.7500	1.7602	1.7657	1.6201	1.6216	1.6256
2.....	4-1/2	.....	.....	2.0000	2.0111	2.0166	1.8557	1.8572	1.8612

TABLE VII. Thread limits, type II, series D through G

Nominal size basic	Pitch				Major diameter		Pitch diameter		
	Series D	Series E	Series F	Series G	Min.	Max.	Basic pitch diameter	Min.	Max.
Mm.	Mm.	Mm.	Mm.	Mm.	Mm.	Mm.	Mm.	Mm.	Mm.
1.5....	.....	.....	.....	0.35	1.524	1.562	1.273	1.278	1.303
2.....	0.45	.....	.....	.....	2.029	2.068	1.708	1.712	1.737
2.5....	.45	.....	.....	.....	2.530	2.568	2.208	2.212	2.238
3.....	.60	.....	.....	.....	3.040	3.091	2.610	2.616	2.654
3.5....	.60	.....	.....	.....	3.541	3.592	3.110	3.117	3.155
4.....	.75	.....	.....	.....	4.056	4.107	3.513	3.526	3.564
4.5....	.75	.....	.....	.....	4.557	4.608	4.013	4.026	4.064
5.....	.90	.....	.....	.....	5.060	5.110	4.415	4.427	4.465
5.5....	.90	.....	.....	.....	5.560	5.611	4.915	4.928	4.966
6.....	1.00	.....	.....	.....	6.066	6.116	5.350	5.364	5.403
7.....	1.00	.....	.....	.....	7.066	7.130	6.350	6.363	6.414
8.....	.....	1.25	.....	.....	8.082	8.146	7.188	7.201	7.252
9.....	.....	1.25	.....	.....	9.081	9.144	8.188	8.202	8.252
10.....	.....	.....	1.00	.....	10.071	10.147	9.350	9.362	9.413
10.....	1.50	.....	.....	.....	10.099	10.175	9.026	9.037	9.101
12.....	.....	1.75	.....	.....	12.111	12.187	10.863	10.876	10.940
14.....	.....	.....	1.25	.....	14.082	14.158	13.188	13.200	13.241
14.....	2.00	.....	.....	.....	14.120	14.196	12.701	12.713	12.776
16.....	2.00	.....	.....	.....	16.119	16.220	14.701	14.714	14.790
18.....	.....	.....	1.50	.....	18.100	18.202	17.026	17.038	17.102
18.....	2.50	.....	.....	.....	18.151	18.252	16.376	16.388	16.464
20.....	2.50	.....	.....	.....	20.163	20.264	18.376	18.402	18.478
22.....	2.50	.....	.....	.....	22.162	22.263	20.376	20.401	20.478
24.....	3.00	.....	.....	.....	24.184	24.285	20.051	22.078	22.154

TABLE VIII. Angle tolerances for series A through C cut thread taps

Threads per inch	Error in half angle	Error in full angle
	Minutes	Minutes
4 and coarser.....	± 30	± 45
4-1/2 to 5-1/2....	± 35	± 53
6 to 9.....	± 40	± 60
10 to 28.....	± 45	± 68
30 to 80.....	± 60	± 90

Note: The nearest thread per inch shall apply for metric thread taps.

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3.18 Workmanship. Workmanship shall be in accordance with the highest grade manufacturing practice now existing for cut thread taps (see 3.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality assurance shall be provided through a quality conformance inspection. The inspection consists of a hardness test, an eccentricity test, a performance test, an examination, and an examination of the preparation of the shipment for delivery. Each sample tap selected shall be inspected for compliance with this specification and MIL-STD-105. The lot represented by the sample shall be rejected when the number of defectives equals or exceeds the rejection number specified in MIL-STD-105 for General Inspection Level II for a given acceptable quality level (AQL). Nonconforming taps shall not be offered for delivery.

4.2.1 Sampling for the quality conformance inspection.

4.2.1.1 Lot. A lot shall consist of all the taps of the same series, style, and size, that are offered for delivery at one time.

4.2.1.2 Sample. The random sample of taps offered for the quality conformance inspection shall be selected from each lot in accordance with the procedures shown in MIL-STD-105 for General Inspection Level II.

4.2.2 Examination. Each of the sample taps selected in accordance with 4.2.1.2 shal be examined to verify compliance with this specification. Examination shall be conducted specified in table IX. Any tap in the sample containing one or more defects shall be rejected, and if the number of defective taps in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The AQL shall be 1.5 percent defective.

TABLE IX. Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Series, style, and size not as specified.
102	Material not as required; not free from cracks, seams, pipe, or scale.
103	Thread form, size, number of threads per inch or pitch (as applicable), and hand of thread not as stipulated.
104	Suitable centers not provided at each end of tap, improperly finished or machined; not internal or external, as required.
105	Thread not a cut thread, not true to shape and well finished; finish of flank and crest galled, chipped, or not smooth, not free from scale, burrs, nicks, cutter and chatter marks or other imperfections which may affect the accuracy of the tap.
106	Flutes not straight, not smoothly ground after hardening, fail to produce an unobstructed chip surface or to effect a refined cutting edge to the crest of the teeth.
107	Number of flutes not as specified; not accurately indexed or depth not sufficient to provide for ample chip clearance.
108	Shank not ground smoothly after hardening; diameter not within the designated tolerance.
109	Evidence of distortion or eccentricity, tap not straight to within the allowable maximum run-out limit.

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TABLE IX. Classification of defects in accordance with MIL-STD-105 (cont.)

Categories	Defects
Major:	
110	Chamfer at threaded end not ground smoothly after hardening with a minimum of chatter marks or not concentric with axis of tap within the required limits.
111	Length of chamfer at threaded end not in accordance with requirements.
112	Length overall, length of thread, length of square, diameter of shank and size of square not within the specified tolerances.
113	Pitch diameter not within the stipulated maximum and minimum limits.
114	Major diameter not in accordance with requirements.
115	Thread lead error and error in half and full angle exceed the permissible tolerances.
116	Tap surfaces not smooth, not free from burrs, raised edges, rust, corrosion, hardening scale, and quenching or grinding checks or cracks.
117	Thread identification markings, missing, incorrect, illegible or not permanent.
Minor:	
201	Marking, manufacturer's name or trademarking missing, incorrect, illegible or not permanent.
202	Preservation, packaging and packing not as required.

4.2.3 Hardness conformance. Each of the sample taps selected in accordance with 4.2.1.2 shall be tested in accordance with 4.3.1 to determine compliance with the hardness requirements specified in paragraph 3.9. The AQL shall be 1.5 percent defective.

4.2.4 Performance. Each sample tap shall be subjected to performance tests at the descretion of the Government. A tap will be defective when it fails to pass the performance test described in paragraph 4.3.3. The AQL shall be 1.5 percent defective.

4.2.5 Examination of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5, or as specified in the contract or purchase order.

4.3 Test procedures. Each sample tap shall be tested for hardness and performance according to the test procedures delineated in paragraphs 4.3.2 and 4.3.3.

4.3.1 Eccentricity test. The total indicator variation (TIV) shall not exceed those values listed in 3.11 when the sample taps are held on centers at their ends and rotated 360°. The AQL shall be 1.5 percent defective.

4.3.2 Hardness test. The shank and thread of the tap shall be tested for proper hardness by using extra fine, Swiss pattern, pillar testing files, cut Nos. 0 and 1, having the proper Rockwell C hardness number. Each tap to be tested shall be mounted in a clamping device. Then, at a predetermined location on the tap, the test file will be applied using short, firm strokes. The test file used shall be in good condition. The tap shall be judged as file hard when the file glides over the tap surface without penetrating the tap surface. When the file bites into the tap surface, it shall be judged as file soft. If for any reason it is decided that a more accurate hardness reading is required, the hardness determination shall be made with appropriate hardness testing equipment with the readings converted, if necessary, to Rockwell "C" scale in accordance with Z76.4 (ASTM E-140).

4.3.2.1 Threaded portion. The threaded portion of the tap shall be tested for Rockwell C61 - 65 hardness by applying a file of Rockwell C60 to the heel of the land. If this portion of the tap tests soft to the file, then the tap shall be defective because it does not comply with the hardness requirements specified in 3.9. A file of Rockwell C67 shall be applied to the heel of the land of the tap. If the tap tests hard to the file, then the tap shall be defective because it does not conform to the requirements of paragraph 3.9. The AQL shall be 1.5 percent defective.

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4.3.2.2 Shank. The shank end of the land shall be tested for a minimum value of Rockwell C45 hardness by applying a file of Rockwell C44 to it. If this portion of the shank tests soft to the file, then the tap shall be judged defective because it does not conform to the requirements specified in 3.9. The AQL shall be 1.5 percent defective.

4.3.3 Performance test. A sharp, centrally ground tap drill of a diameter that will produce approximately 75 percent of full thread for the taps selected for testing shall be used. Eight tap holes shall be drilled for each tap under test, in a mild-steel plate of a thickness approximating the diameter of the tap used. Using a suitable coolant, the three holes shall be tapped. First and eighth hole shall be spot checked with a class 2B and a class 2 (HI) thread plug gage to determine the uniformity of the holes produced. If any difference exists between the first and third hole, the tap used shall be rechecked for dullness, breakage or other detrimental failure in the tap. If tapped holes appear uniform when checked with gage and the tap appears sharp and still in good condition, it has passed the test.

4.4 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5, or as specified in the contract or purchase order.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40.

6. NOTES

6.1 Intended use. The taps listed herein are carbon steel, cut-thread taps used for cutting internal bolt or screw threads. These carbon steel taps are used principally for hand tapping.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, series, and style of taps (see 1.2).
(c) Size and threads per inch required (see applicable tables).
(d) If left-hand taps are required (see 3.6).

6.3 Replacement information. Styles a, b, and c taps have been reclassified as styles 1, 2, and 3 taps, respectively and are completely interchangeable.

MILITARY INTEREST:

User activities:

Army - GL, WC
Navy - MC, SH
Air Force - 85

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST:

GSA-FSS

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